

Mesporin®-1000 I.M./I.V.

Mesporin®-250/500 I.M.

Long-acting, parenteral, broad-spectrum
bactericidal antibiotic

Composition

Mesporin®-1000 I.V.

Each vial contains:

Ceftriaxone 1000 mg in the form of disodium salt.

Each ampoule of solvent contains 10 ml of sterile
water for injection.

Mesporin®-1000 I.M.

Each vial contains:

Ceftriaxone 1000 mg in the form of disodium salt.

Each ampoule of solvent contains 3.5 ml of lidocaine
HCl 1%.

Mesporin®-500 I.M.

Each vial contains:

Ceftriaxone 500 mg in the form of disodium salt.

Each ampoule of solvent contains 2 ml of lidocaine
HCl 1%.

Mesporin®-250 I.M.

Each vial contains:

Ceftriaxone 250 mg in the form of disodium salt.

Each ampoule of solvent contains 25 ml of lidocaine
HCl 1%.

Mesporin® contains approximately 83 mg (3.6 mEq)
of sodium per gram of ceftriaxone.

Properties, effects

The bactericidal activity of ceftriaxone results from
inhibition of cell wall synthesis. Ceftriaxone exerts *in
vitro* activity against a wide range of gram-negative
and gram positive microorganisms. Ceftriaxone is
highly stable to most β -lactamases, both penicilli-
nases and cephalosporinases, of gram-negative and
gram-positive bacteria. Ceftriaxone is usually active
against the following microorganisms *in vitro* and in
clinical infections:

Gram-positive aerobes	median values	
	MIC ₅₀ * (in mg/l)	MIC ₉₀ ** (in mg/l)
<i>Staphylococcus aureus</i> (methicillin-sensitive)	4	4
<i>Staphylococci</i> , Coagulase-negative	4	16
<i>Streptococcus pyogenes</i> (beta-haemolytic, group A)	0.03	0.03
<i>Streptococcus agalactiae</i> (beta-haemolytic, group B)	≤0.06	0.06
<i>Streptococci</i> , beta-haemolytic (not group A or B)	≤0.06	0.06
<i>Streptococcus viridans</i>	0.125	0.5
<i>Streptococcus pneumoniae</i>	≤0.06	0.06

* MIC₅₀ = minimal inhibitory concentration for 50%
of the tested strains.

**MIC₉₀ = minimal inhibitory concentration for 90%
of the tested strains.

Gram-negative aerobes	median values	
	MIC ₅₀ (in mg/l)	MIC ₉₀ (in mg/l)
<i>Acinetobacter Iwoffi</i>	2	8
<i>Acinetobacter anitratus</i> ¹⁾ (mainly <i>A. baumannii</i>)	8	32
<i>Aeromonas hydrophila</i>	0.25	4
<i>Alcaligenes faecalis</i>	1	8
<i>Alcaligenes odorans</i>	≤0.25	0.5
<i>Alcaligenes-like bacteria</i>	≤0.25	0.5
<i>Borrelia burgdorferi</i>	≤0.06	≤0.06
<i>Burkholderia cepacia</i>	2	16
<i>Capnocytophaga</i> spp.	≤0.06	4
<i>Citrobacter diversus</i> (including <i>C. amalonaticus</i>)	0.125	0.125
<i>Citrobacter freundii</i> ¹⁾	0.125	16

Gram-negative aerobes	median values	
	MIC ₅₀ (in mg/l)	MIC ₉₀ (in mg/l)
<i>Escherichia coli</i>	≤0.06	0.125
<i>Enterobacter aerogenes</i> ¹⁾	2	16
<i>Enterobacter cloacae</i> ¹⁾	0.5	16
<i>Enterobacter</i> spp. (other) ¹⁾	0.25	32
<i>Haemophilus ducreyi</i>	0.004	0.004
<i>Haemophilus influenzae</i>	≤0.008	0.06
<i>Haemophilus parainfluenzae</i>	0.016	0.06
<i>Hafnia alvei</i>	0.125	2
<i>Klebsiella oxytoca</i>	≤0.06	0.125
<i>Klebsiella pneumoniae</i> ²⁾	≤0.06	0.125
<i>Moraxella catarrhalis</i> (formerly <i>Branhamella catarr.</i>)	0.125	0.5
<i>Moraxella osloensis</i>	≤0.25	≤0.25
<i>Moraxella</i> spp. (other)	≤0.25	≤0.25
<i>Morganella morganii</i>	0.06	1
<i>Neisseria gonorrhoeae</i>	≤0.008	0.06
<i>Neisseria meningitidis</i>	≤0.008	0.008
<i>Pasteurella multocida</i>	≤0.06	0.06
<i>Plesiomonas shigelloides</i>	≤0.06	0.06
<i>Proteus mirabilis</i>	≤0.06	0.06

<i>Proteus penneri</i> ¹⁾	1	64
<i>Proteus vulgaris</i>	≤0.06	2
<i>Pseudomonas fluorescens</i> ¹⁾	16	64
<i>Pseudomonas</i> spp. (other) ¹⁾	8	16
<i>Providencia rettgeri</i>	≤0.06	2
<i>Providencia</i> spp. (other)	≤0.06	0.5
<i>Salmonella typhi</i>	≤0.06	0.125
<i>Salmonella</i> spp. (<i>Enteritidis</i> group)	≤0.06	0.06
<i>Serratia marcescens</i>	0.5	2
<i>Serratia</i> spp. (other)	0.25	16
<i>Shigella</i> spp.	0.03	0.25
<i>Vibrio</i> spp.	≤0.06	0.25
<i>Yersinia enterocolitica</i>	≤0.125	0.125
<i>Yersinia</i> spp. (other)	0.25	2

Anaerobes	median values	
	MIC ₅₀ (in mg/l)	MIC ₉₀ (in mg/l)
<i>Bacteroides</i> spp. ³⁾ (bile-sensitive)	2	16
<i>Clostridium</i> spp. (without <i>C. perfringens</i> group)	2	16
<i>Fusobacterium nucleatum</i>	1	2
<i>Fusobacterium</i> spp. (remaining)	0.125	0.25
<i>Gaffkia anaerobica</i> (formerly <i>Peptococcus</i>)	0.125	1
<i>Peptostreptococci</i>	0.125	1

Susceptibility tests

Susceptibility to ceftriaxone can be determined by
the disk diffusion test or by the agar or broth dilution
test using standardised techniques for susceptibility
testing such as those recommended by the National
Committee for Clinical Laboratory Standards
(NCCLS). The NCCLS issued the following interpre-
tative breakpoints for ceftriaxone:

	Susceptible	Moderately Susceptible	Resistant
<i>Dilution test.</i> inhibitory concen- trations (mg/l)	≤8	16-32	≥64
Diffusion test (disk with 30 ng ceftriaxone), inhibition zone diameter in mm	≥21	20-14	≤13

Microorganisms should be tested with the ceftriax-
one disk since it has been shown by *in vitro* tests to
be active against certain strains resistant to cephalo-
sporin class disks.

Where NCCLS recommendations are not in daily
use, alternative, well standardised, susceptibility
interpretative guidelines such as those issued by
DIN, ICS and others may be substituted.

Resistances

- 1) Some isolates of these species are resistant to ceftriaxone because of the derepression of the chromosomal β lactamase.
- 2) Some isolates of *Klebsiella pneumoniae* are resistant to ceftriaxone because of the plasmid-dependent β -lactamase production.
- 3) Some isolates of *Bacteroides* spp. are resistant to ceftriaxone.

Many strains of the β -lactamase producing *Bacteroides* spp. (specifically *B. fragilis*) are resistant.

Clostridium difficile is resistant.

Methicillin-resistant *Staphylococcus* spp. are resistant to cephalosporins including ceftriaxone. *Enterococcus faecalis* and *Enterococcus faecium* as well as *Listeria monocytogenes* are resistant as a rule.

Many strains of gram-negative aerobes demonstrating multiple resistance to other antibiotics such as amino- and ureido-penicillins, older cephalosporins and aminoglycosides are sensitive to ceftriaxone. *Treponema pallidum* is sensitive *in vitro* and in animal experiments. Clinical trials have shown that primary and secondary syphilis responds well to treatment with ceftriaxone.

With few exceptions, clinical isolates of *Pseudomonas aeruginosa* are resistant to ceftriaxone.

Pharmacokinetics

Ceftriaxone displays non-linear pharmacokinetics. All pharmacokinetic parameters, with the exception the elimination half-life, are dose-dependent if they refer to the overall concentration (free and protein-bound ceftriaxone).

Absorption

After an i.m. injection of 1 g ceftriaxone, a maximum plasma concentration of 81 mg/l was achieved after

2-3 hours. After a single i.v. infusion of 1 g, a concentration of 168.1 ± 28.2 mg/l was achieved after 30 minutes. After a single i.v. infusion of 2g a concentration of 256.9 ± 16.8 mg/l was achieved after 30 minutes.

The areas under the plasma concentration-time curves after i.v. and i.m. administration are identical. This means that the bioavailability of ceftriaxone administered i.m. is 100%.

Distribution

The distribution volume lies between 7 and 12 l.

With intravenous administration, ceftriaxone rapidly penetrates interstitial body fluids, where bactericidal concentrations remain active against sensitive pathogens for 24 hours.

After a dose of 1-2g, ceftriaxone shows good penetration into tissue and body fluids; concentrations above the minimal inhibitory concentration for most pathogens were measured for more than 24 hours in over 60 tissue and body fluids, incl. lung, heart, biliary tract, liver, middle ear, nasal mucous membrane and bones as well as cerebrospinal, pleural, synovial and prostate fluid.

Ceftriaxone is reversibly bound to plasma albumin, and the binding decreases with the increase in the concentration, e.g. 95% binding at plasma concentrations of <100 mg/l to 85% binding at a plasma concentration of 300 mg/l. Owing to the lower albumin content, the proportion of free ceftriaxone in the interstitial fluid is correspondingly higher than in plasma.

Ceftriaxone penetrates the inflamed meninges of neonates, infants and children. An average maximal concentration of 18 mg/l in the CSF is reached approx. 4 hours after i.v. application of ceftriaxone 50-100 mg/kg. The average extent of diffusion into the CSF in bacterial meningitis is about 17% of the plasma concentration; in aseptic meningitis 4%. Twenty-four hours after injection of ceftriaxone in doses of 50-100 mg/kg body weight, concentrations of >1.4 mg ceftriaxone per liter have been found in the CSF. In adult patients with meningitis, administration of 50 mg/kg leads within 2-24 hours to CSF

concentrations which are several times higher than the required minimal inhibitory concentration against the most common causative organisms in meningitis. Ceftriaxone penetrates the placental barrier. Ceftriaxone is excreted into breast milk in low concentrations (3-4% of the mother's plasma concentration after 4-6 hours).

Metabolism

Ceftriaxone is not metabolized in the organism itself. Bowel flora converts the active agent into inactive metabolites after biliary excretion into the lumen of the bowel.

Elimination

Plasma clearance is 10-22 ml/min. Renal clearance is 5-12 ml/min. In adults, 50-60% of ceftriaxone is excreted in the urine as unchanged drug while 40-50% is excreted unchanged in the bile. The serum elimination half-life in healthy subjects is about eight hours.

Kinetics in special clinical situations

In neonates, renal elimination accounts for about 70% of the dose. In infants less than eight days old and subjects over 75 years of age the average elimination half-life is about 2-3 times as long as in healthy young adults. The pharmacokinetics of ceftriaxone are only minimally altered in patients with renal impairment or hepatic dysfunction, and the elimination half-life is only slightly increased. If kidney function alone is impaired, biliary elimination of ceftriaxone is increased; hepatic dysfunction alone leads to increased renal excretion.

Indications

Mesporin is indicated for the treatment of infections caused by strains susceptible to ceftriaxone, in the conditions listed below:

- respiratory tract infections, particularly pneumonia and ear, nose and throat infections;
- abdominal infections (peritonitis, infections of the biliary tract and the gastrointestinal tract);
- renal- and urinary tract infections;
- infections of the genital tract and organs, including gonorrhea;
- sepsis;

- infections of the bones, joints, soft tissues, skin and wounds;
- infections in immunocompromised patients;
- meningitis;
- disseminated Lyme borreliosis (Stage II and III);
- perioperative infection prophylaxis in operations of the GI tract, biliary tract, in urogenital and gynecological procedures - however, only in case of potential or established contamination.

Dosage/Use

Recommended dosage

Adults and children over twelve years: The usual dosage is 1-2 g of ceftriaxone once daily (every 24 hours). In severe infections, or infections caused by moderately susceptible pathogens, the daily single dose may be raised to 4 g.

Neonates, infants and children up to twelve years: The following dosage schedules are recommended for once daily administration:

Neonates (up to two weeks): A daily dose of 20-50 mg/kg body weight; not to exceed 50 mg/kg. It is not necessary to differentiate between premature babies and infants born at term.

Infants and children (two weeks to 12 years): A daily dose of 20-80 mg/kg.

For children with a body weight of 50 kg or more the usual adult dosages should be used. Intravenous doses of 50 mg/kg/body weight or higher should be given by infusion over at least 30 minutes.

Elderly patients: The recommended dosages for adults do not require modification.

Duration of treatment: Duration of treatment varies according to the indication and the course of the illness.

Combination therapy: Synergy between aminoglycosides and ceftriaxone has been demonstrated experimentally with many gram-negative bacteria. Although enhanced activity of such combinations is not always predictable, it should be considered in severe, life-threatening infections caused by pathogens such as *Pseudomonas aeruginosa*. Because of physical incompatibility the two drugs must be administered separately at the recommended dosage.

Special dosage instructions

Meningitis:

In bacterial meningitis in infants and children, the treatment begins with doses of 100 mg/kg (not to exceed 4 g) once daily. As soon as the causative organism has been identified and its sensitivity determined, the dosage can be reduced accordingly. The best results have been found with the following duration of treatment:

<i>Neisseria meningitidis</i>	4 days
<i>Haemophilus influenzae</i>	6 days
<i>Streptococcus pneumoniae</i>	7 days

Lyme borreliosis:

The dosage for the treatment of Lyme borreliosis in children and adults is 50 mg/kg to maximum 2g once daily during 14 days.

Gonorrhoea:

For the treatment of gonorrhoea (penicillinase-producing and non-penicillinase-producing strains) the single i.m. administration of 250 mg ceftriaxone is recommended.

Perioperative prophylaxis:

To prevent postoperative infections in contaminated or potentially contaminated surgical procedures, it is recommended, depending on risk of infection, to administer a single dose of 1-2g of ceftriaxone 30 to 90 minutes prior to surgery. In colorectal surgery the simultaneous administration of ceftriaxone and a 5-nitroimidazole, e.g. ornidazole is recommended.

Impaired renal and liver function:

In patients with impaired renal function there is no need to reduce the dosage of ceftriaxone, provided liver function is intact. Only in cases of preterminal renal failure (creatinine clearance <10 ml/min) the daily dose of ceftriaxone must not exceed 2g. In patients with impaired liver function there is no need to reduce the dose of ceftriaxone, provided renal function is intact. In cases of concomitant severe renal and liver dysfunction, the plasma concentration of ceftriaxone should be determined at regular intervals. Dosage adjustment may be required as these

patients can show a reduced elimination of ceftriaxone. In patients undergoing dialysis no supplementary dosing is required after dialysis. More important, the serum concentration should be monitored to determine if dosage adjustments are necessary, as these patients may have a reduced elimination rate. A daily dose of 2g should not be exceeded in patients undergoing dialysis.

Directions for use

The physical and chemical stability of reconstituted solutions is retained for six hours at room temperature (or 24 hours at 5°C). However, as a general rule, the solutions should be used immediately after preparation. Depending on concentration, solvent used, and the length of storage, the color of the solutions ranges from light yellow to amber. This property of the active ingredient has no significance for the efficacy or tolerance of the drug.

Intramuscular injection:

For i.m. injection, Mesporin should be dissolved in the appropriate volume of solvent as follows:

a) 250mg or 500mg of ceftriaxone in 2ml of 1% lidocaine solution; b) 1g of ceftriaxone in 3.5 ml of 1% lidocaine solution; and administered by deep intramuscular injection. It is recommended that not more than 1g of ceftriaxone should be injected on either side. The lidocaine solution must never be administered intravenously.

Intravenous injection:

For i.v. injection, 1 g of ceftriaxone is dissolved in 10 ml of sterile water for injection and administered by slow intravenous injection lasting two to four minutes.

Intravenous infusion:

The infusion should last at least 30 minutes. 2g Mesporin are dissolved in 40 ml of one of the following calcium-free infusion solutions for the i.v. infusion: sodium chloride 0.9%, sodium chloride 0.45% + glucose 2.5%, glucose 5%, glucose 10%, dextran 6% in glucose 5%, hydroxyethyl starch 6-10%, water for injection purposes. Because of possible incompatibility, solutions containing Mesporin must not be mixed with or joined to solutions containing other

antibiotics, and must not be added to infusion solutions other than those listed above.

However, 2g ceftriaxone and 1 g ornidazole are physically and chemically compatible in 250 ml physiological sodium chloride or glucose solution.

Do not use diluents containing calcium, such as Ringer's Solution or Hartmann's Solution, to reconstitute ceftriaxone. Particulate formation can result.

Contraindications

Ceftriaxone is *contraindicated* in patients with known allergy to the cephalosporin class of antibiotics. This product should be given cautiously to penicillin-sensitive patients; the possibility of allergic cross-reactions should be borne in mind.

Warnings and precautions

Even after a detailed medical history has been established, an anaphylactic reaction cannot be excluded. If allergic reactions occur, Mesporin must be discontinued immediately and suitable therapy initiated.

Ceftriaxone can prolong prothrombin time. In case of a suspected vitamin K deficit, a check of the Quick value is therefore recommended.

In the case of severe or lasting diarrhoea, pseudomembranous colitis, caused by antibiotics and potentially life-threatening, must be taken into consideration. Therefore, in this case, Mesporin must be discontinued immediately and suitable therapy initiated.

In the case of long-term use of Mesporin, non-sensitive pathogens may supervene. Therefore, close monitoring of the patient is essential. If a super infection occurs during treatment, appropriate measure must be taken.

During ultrasound investigations of the gall bladder, shadows can mistakenly be interpreted as gallstones. Such shadows are generally precipitation of the calcium salt of ceftriaxone. This precipitation has usually occurred when doses higher than those recommended were administered. When treatment with Mesporin ends or is interrupted, these shadows disappear.

These findings were rarely accompanied by symptoms. In cases where symptoms occur, conservative,

non-surgical treatment is recommended. The doctor should decide whether to discontinue Mesporin in patients with symptoms.

In rare cases, there have been reports of pancreatitis possibly caused by cholestasis in patients treated with Mesporin. Most patients showed risk factors for cholestasis and bile slime in consultation, for example extensive previous treatment, serious illness or completely parenteral diet. It cannot be excluded that the precipitation stemming from Mesporin in the gall bladder can be a trigger or a co-factor.

Ceftriaxone can displace bilirubin from its binding to serum albumin. Care is therefore necessary when treating new-born babies with hyperbilirubinaemia. Newborn and in particular prematurely born patients with the risk of bilirubin encephalopathy should not be treated with Mesporin.

The blood picture should be checked regularly during long-term treatment.

Care must be taken with patients with impaired renal function in the case of concomitant treatment with aminoglycosides and diuretics.

Ceftriaxone must not be mixed or administered simultaneously with calcium-containing solutions or products, even via different infusion lines.

Calcium containing solutions or products must not be administered within 48 hours of last administration of ceftriaxone.

Cases of fatal reactions with calcium-ceftriaxone precipitates in lungs and kidneys in both term and premature neonates have been described. In some cases the infusion lines and times of administration and calcium-containing solutions differed.

Pregnancy/lactation

Pregnancy category B. Ceftriaxone penetrates the placental barrier (see <<Pharmacokinetics: distribution>>). Although neither mutagenic nor teratogenic characteristics have been established in the corresponding pre-clinical examinations, Mesporin should only be used in the case of imperative need during pregnancy, particularly in the first three months.

Because ceftriaxone is excreted into breast milk - even if only in low concentrations - care must be taken in treating lactating women with Mesporin.

Undesirable effects

During treatment with ceftriaxone the following adverse reactions were observed, which disappeared either spontaneously or after withdrawal of treatment.

Systemic tolerance

Gastrointestinal complaints (about 2% of cases): loose stools or diarrhoea, nausea, vomiting, stomatitis, glossitis.

Haematological changes (about 2%): eosinophilia, leukopenia, granulocytopenia, haemolytic anaemia, thrombocytopenia, prolongation of the prothrombin time.

Isolated cases of agranulocytosis ($< 500/\text{mm}^3$) have been observed, most of them after total doses of 20 g or more.

Skin reactions (about 1-2%)

Exanthem, allergic dermatitis, pruritus, urticaria, oedema.

There have been reports of isolated cases of severe skin reactions (erythema multiforme, Stevens-Johnson syndrome or Lyell's syndrome/toxic epiderma necrolysis).

Other side effects: Headache and dizziness, increased liver enzymes, oliguria, increase in serum creatinine, mycosis of the genital tract, fever, shivering, and anaphylactic or anaphylactoid reactions, as well as precipitation of the calcium salt of ceftriaxone in the gall bladder.

Pseudomembranous enterocolitis and prolongation of the prothrombin time have been reported as very rare side effects.

Precipitation in the kidneys has been very rarely observed, and usually in children over 3 years of age who were treated with either high daily doses (for example $\geq 80 \text{ mg/kg/day}$) or overall doses in excess of 10g and showing other risk factors (e.g. reduced fluid intake, confinement to bed, etc.). This side effect can occur with or without symptoms, can lead to renal insufficiency and is reversible after discontinuation of Mesporin.

Local side effects

Phlebitis less than 1% after i.v. administration. This can be avoided by slow intravenous injection (two to four minutes). Intramuscular injection without lido-

caine solution is painful.

Interactions

After simultaneous administration of high doses of ceftriaxone and potent diuretics, such as furosemide, no disturbance of renal function has been observed. A <<disulfiram-like>> effect after ceftriaxone administration and concomitant intake of alcoholic beverages has not been demonstrated. Ceftriaxone does not contain an N-methylthiotetrazole moiety that may cause ethanol intolerance and bleeding problems, as is the case with some other cephalosporins. Probenecid has no influence on the elimination of ceftriaxone.

There is no proof that ceftriaxone increases the renal toxicity of aminoglycosides.

Bacteriostatic agents can interfere unfavourably with the bactericidal effect of cephalosporins.

Antagonistic effects have been observed in an in vitro study of ceftriaxone in combination with chloramphenicol.

Overdosage

Excessively high plasma concentrations of ceftriaxone cannot be reduced through haemo- or peritoneal dialysis. Symptomatic measures are recommended for treating patients after overdosage.

Further remarks

Incompatibilities

Mesporin must not be mixed with solutions containing calcium such as Hartmann's or Ringer's solution.

Ceftriaxone is incompatible with amsacrine, vancomycin, fluconazole and aminoglycosides.

Influence on diagnostic methods

In rare cases under treatment with Mesporin the Coombs test can give a false positive result. Tests for galactosemia can give a false positive result under Mesporin as is the case with other antibiotics. Non-enzymatic methods for determining urinary sugar can also give a false positive result. For this reason, urinary sugar should be determined enzymatically during therapy with Mesporin.

Presentation

Mesporin®-1000 I.V.

Packings for 1 i.v. injection containing:

1 vial with dry substance equivalent to 1000 mg of ceftriaxone, and 1 ampoule with 10 ml of sterile water for injection.

Mesporin®-1000 I.M.

Packings for 1 i.m. injection containing:

1 vial with dry substance equivalent to 1000 mg of ceftriaxone, and 1 ampoule with 3.5 ml of 1% lidocaine HCl solution.

Mesporin®-500 I.M.

Packings for 1 i.m. injection containing:

1 vial with dry substance equivalent to 500 mg of ceftriaxone, and 1 ampoule with 2 ml of 1% lidocaine HCl solution.

Mesporin®-250 I.M.

Packings for 1 i.m. injection containing:

1 vial with dry substance equivalent to 250 mg of ceftriaxone, and 1 ampoule with 2 ml of 1% lidocaine HCl solution.

Storage conditions

Store dry below 30°C.

Reconstituted solutions are stable for 6 hours at room temperature in daylight, or 24 hours at 5°C.

Other information

Shelf life: 3 years

Expiry: The preparation should not be used after the date marked "EXP" on the pack.

Note: Medicines should be kept out of reach of children.

Information date

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